USC University of Southern California

ECE 105: Introduction to Electrical Engineering

Lecture 10

Device 2

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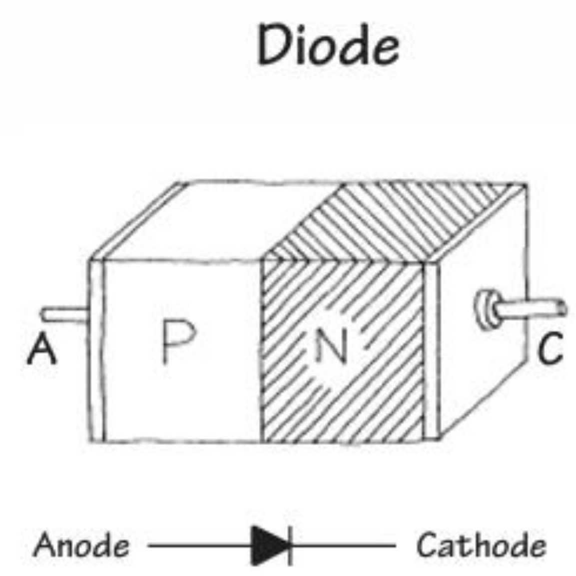
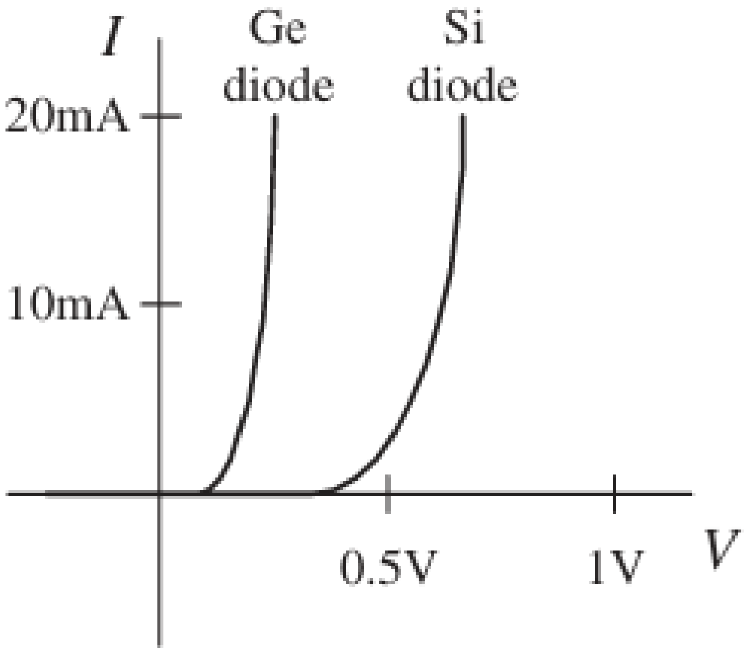
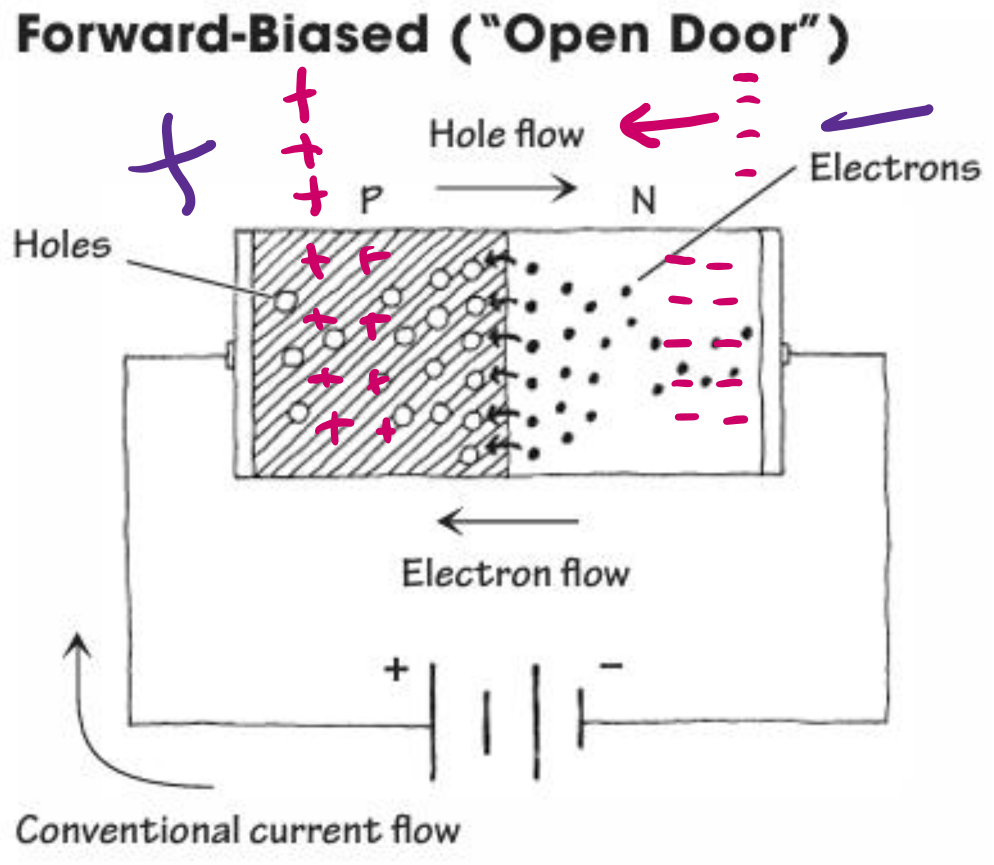
What is a diode

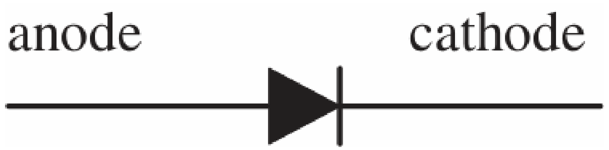




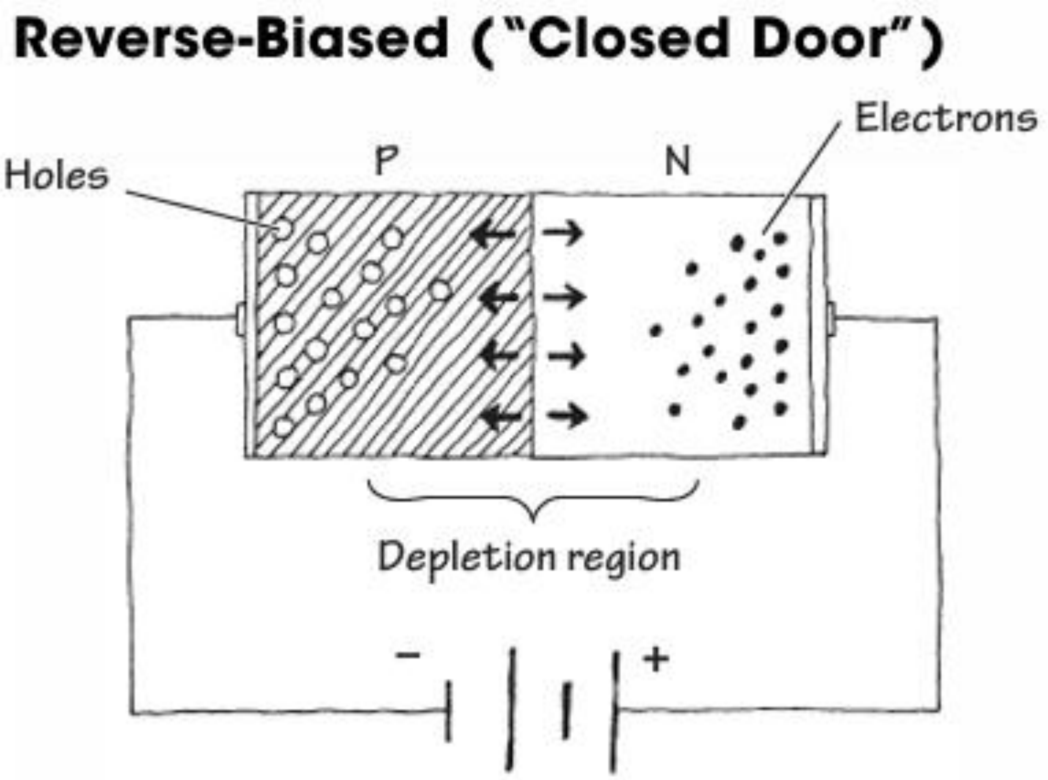
- A diode is a device that exhibits ‘rectifying’ behavior
- Can be made from a physical junction between an n-type semiconductor and a p-type semiconductor
- Can also be made from a physical junction between a metal and semiconductor (not discussed here)

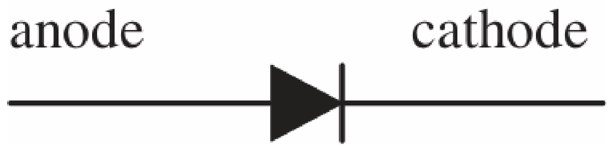
Forward biasing a diode



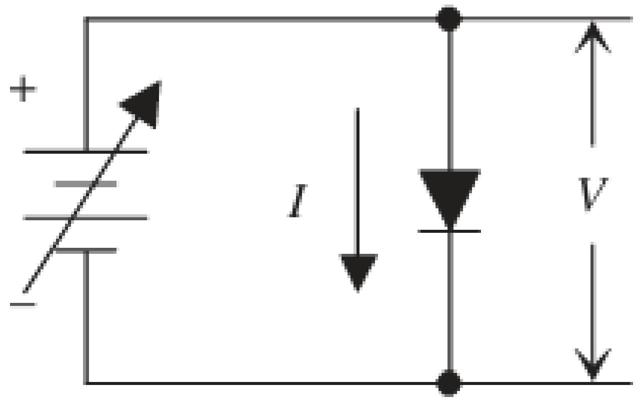


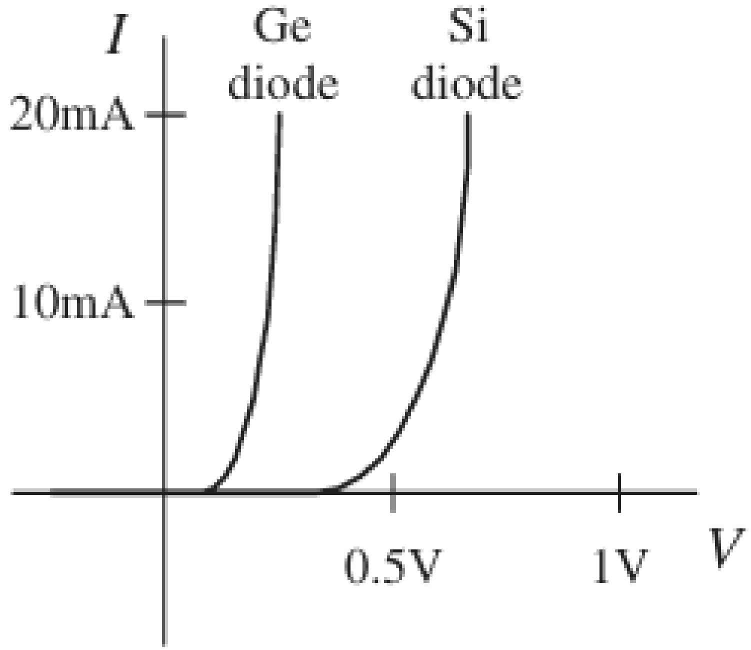
Reverse biasing a diode



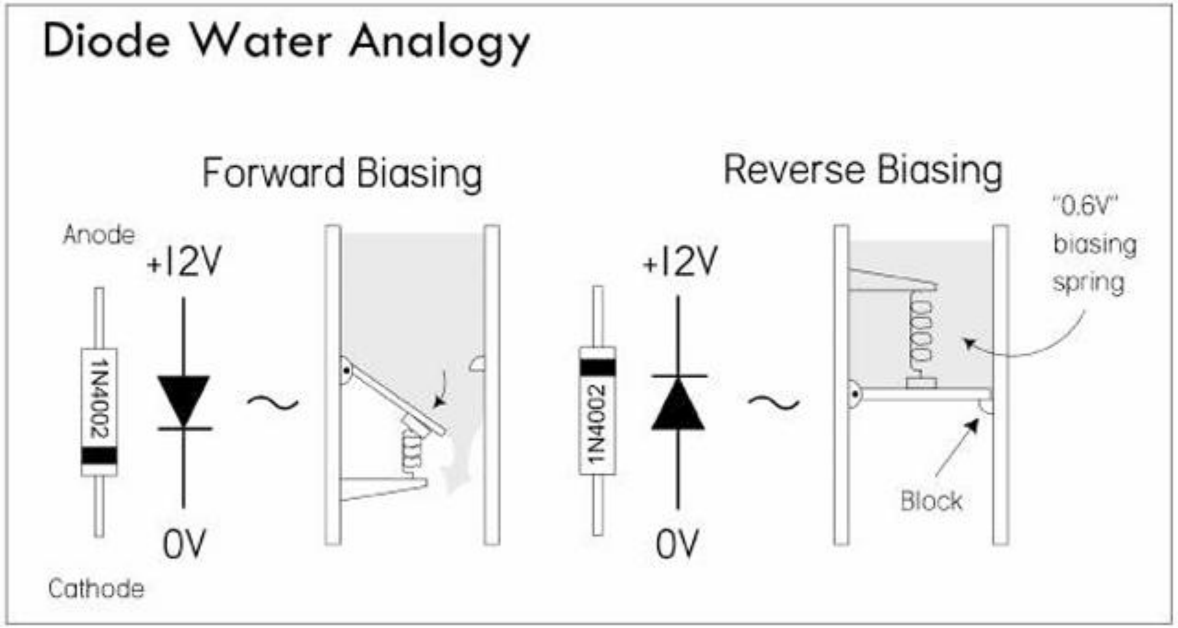


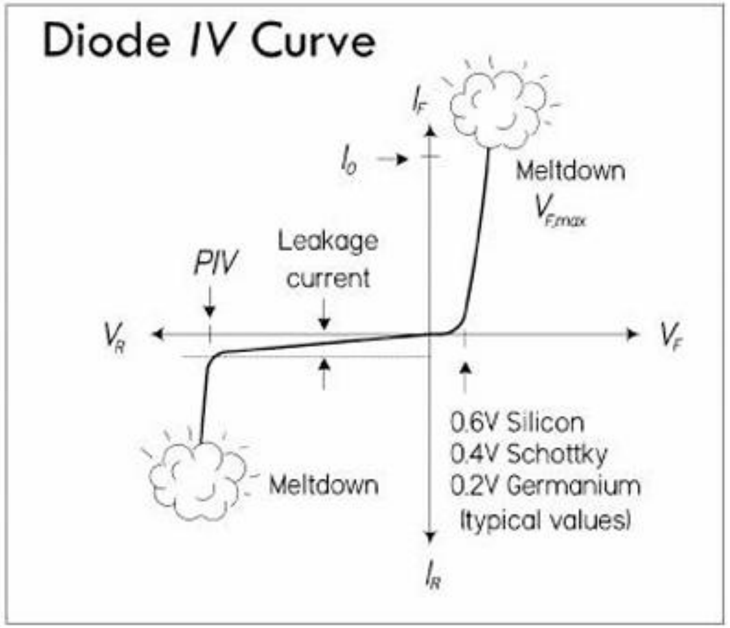
Diode characterization





Water analogy





Diode Current Equation

$$I = I_0\left(e^{\frac{qV}{nkT}} - 1\right)$$

- This equation partially describes the behavior of a non-ideal diode

Diode Current Equation

$$I = I_s \left(e^{\frac{V}{nV_T}} - 1 \right)$$

where:

- I = diode current
- I_s = reverse saturation current (very small, e.g. 10^{-12} A)
- V = applied voltage across the diode
- n = ideality factor (≈ 1 for silicon)
- $V_T = \frac{kT}{q} \approx 25.9\,\text{mV at 300K}$

Example 1 — Forward bias at 0.7 V

- Let $I_s = 10^{-12}\,\text{A}$, $n = 1$, $V_T = 0.026\,\text{V}$
- $V = 0.7\,\text{V}$

$$I = 10^{-12} \left(e^{\frac{0.7}{0.026}} - 1 \right)$$

$$\frac{0.7}{0.026} \approx 26.9 \quad \Rightarrow \quad e^{26.9} \approx 5.3 \times 10^{11}$$

$$I \approx 10^{-12} \times 5.3 \times 10^{11} \approx 0.53\,\text{A}$$

$$I = I_s \left(e^{\frac{V}{nV_T}} - 1 \right)$$

$$I_s = 10^{-12} \left(e^{\frac{.7}{.026}} - 1 \right)$$

Diode Current Equation

Example 2 — Forward bias at 0.6 V

$$I = 10^{-12}\left(e^{\frac{0.6}{0.026}} - 1\right)$$

$$\frac{0.6}{0.026} \approx 23.1, \quad e^{23.1} \approx 1.1 \times 10^{10}$$

$$I \approx 10^{-12} \times 1.1 \times 10^{10} = 0.011\,\mathrm{A} = 11\,\mathrm{mA}$$

Example 3 — Reverse bias at -5 V

$$I = 10^{-12}\left(e^{\frac{-5}{0.026}} - 1\right)$$

$$\frac{-5}{0.026} \approx -192 \quad \Rightarrow \quad e^{-192} \approx 0$$

$$I \approx -10^{-12}\,\mathrm{A}$$

Example 4 — Small voltage 0.1 V

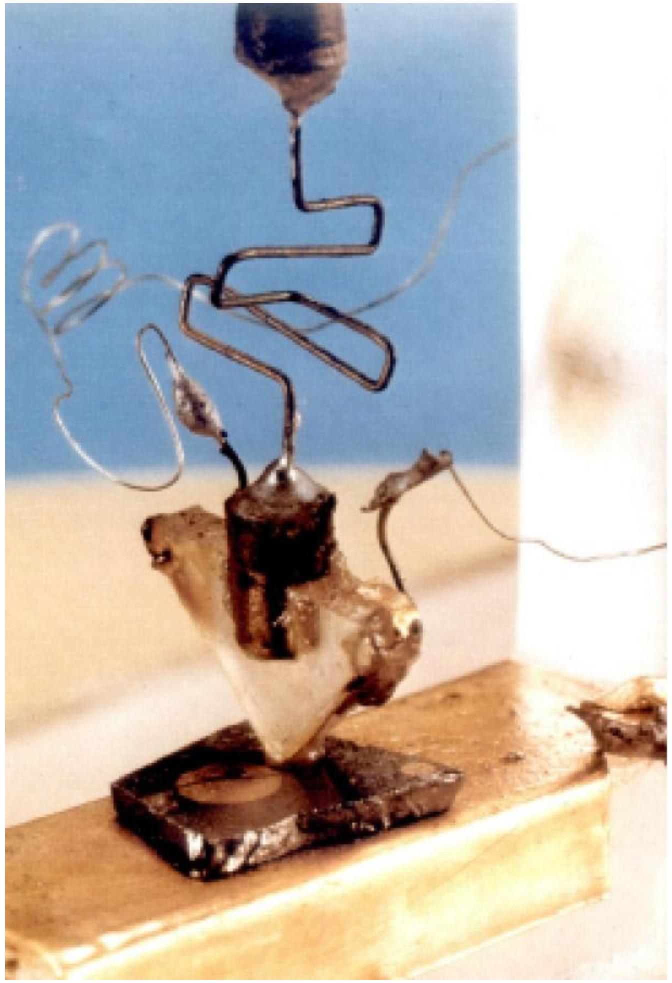
$$I = 10^{-12}\left(e^{\frac{0.1}{0.026}} - 1\right)$$

$$\frac{0.1}{0.026} \approx 3.85, \quad e^{3.85} \approx 47$$

$$I \approx 10^{-12} \times (47 - 1) \approx 4.6 \times 10^{-11}\,\mathrm{A}$$

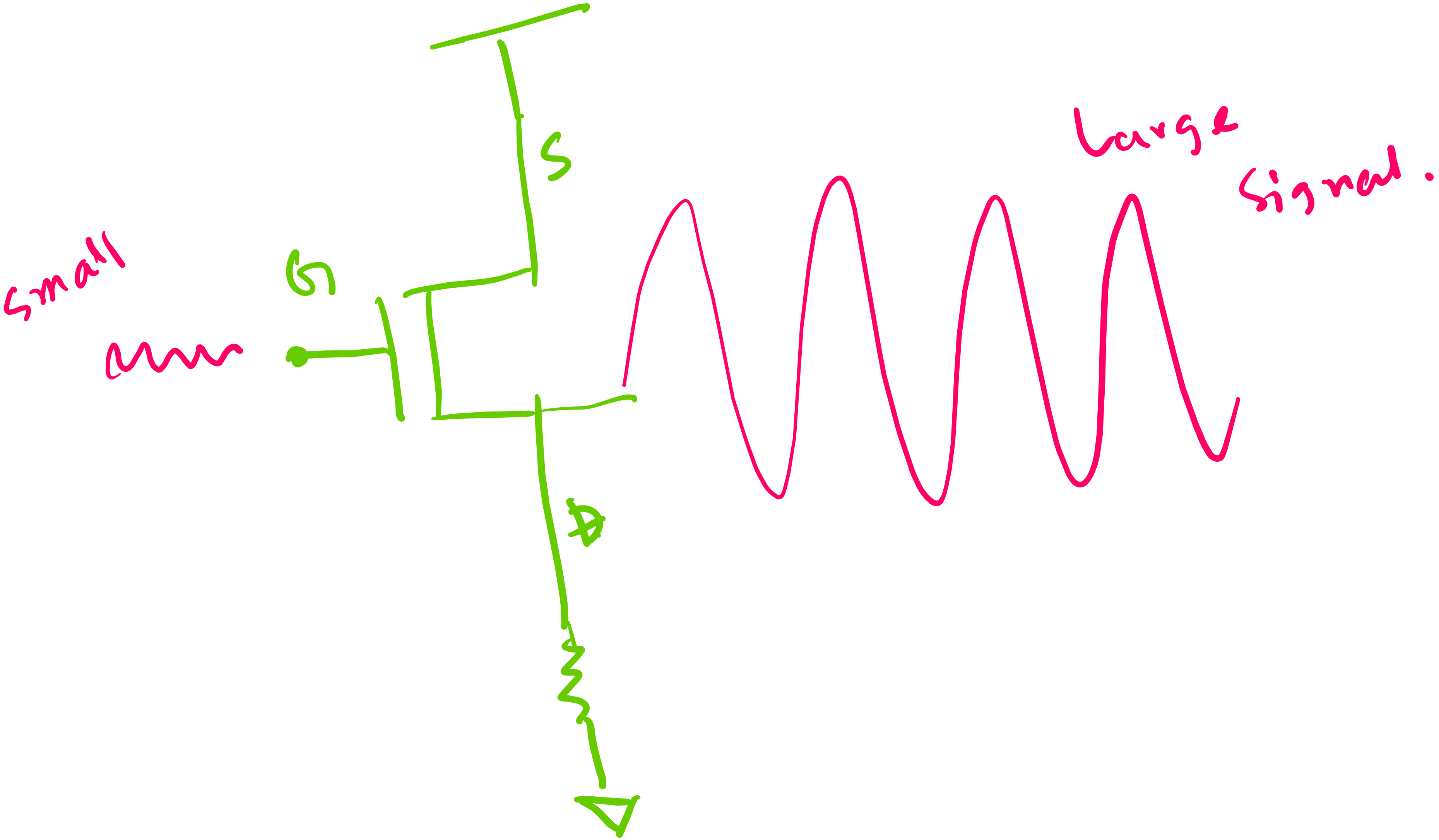
The First Transistor

•At **Bell Telephone Laboratories**, research was focused on finding a solid-state amplifier for telephone signal processing.
•The three key scientists were:
•**John Bardeen** (theorist)
•**Walter Brattain** (experimentalist)
•**William Shockley** (team leader, physicist)



•**December 16, 1947**: John Bardeen and Walter Brattain successfully demonstrated the **first working transistor** at Bell Labs.
•It was a **point-contact transistor** made of a small slab of germanium with two closely spaced gold contacts.
•When input current was applied to one contact, it modulated the conductivity of the germanium, amplifying the signal seen at the other contact.
Key outcome: It could amplify electrical signals like a vacuum tube, but in a device only a few millimeters wide.

Transistor amplification



1956 Physics Nobel Prize

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"for their researches on semiconductors and their discovery of the transistor effect"

William Bradford Shockley

1/3 of the prize

USA

Semiconductor Laboratory of Beckman Instruments, Inc.
Mountain View, CA, USA

John Bardeen

1/3 of the prize

USA

University of Illinois
Urbana, IL, USA

Walter Houser Brattain

1/3 of the prize

USA

Bell Telephone Laboratories
Murray Hill, NJ, USA

Invention of the Field-Effect Transistor

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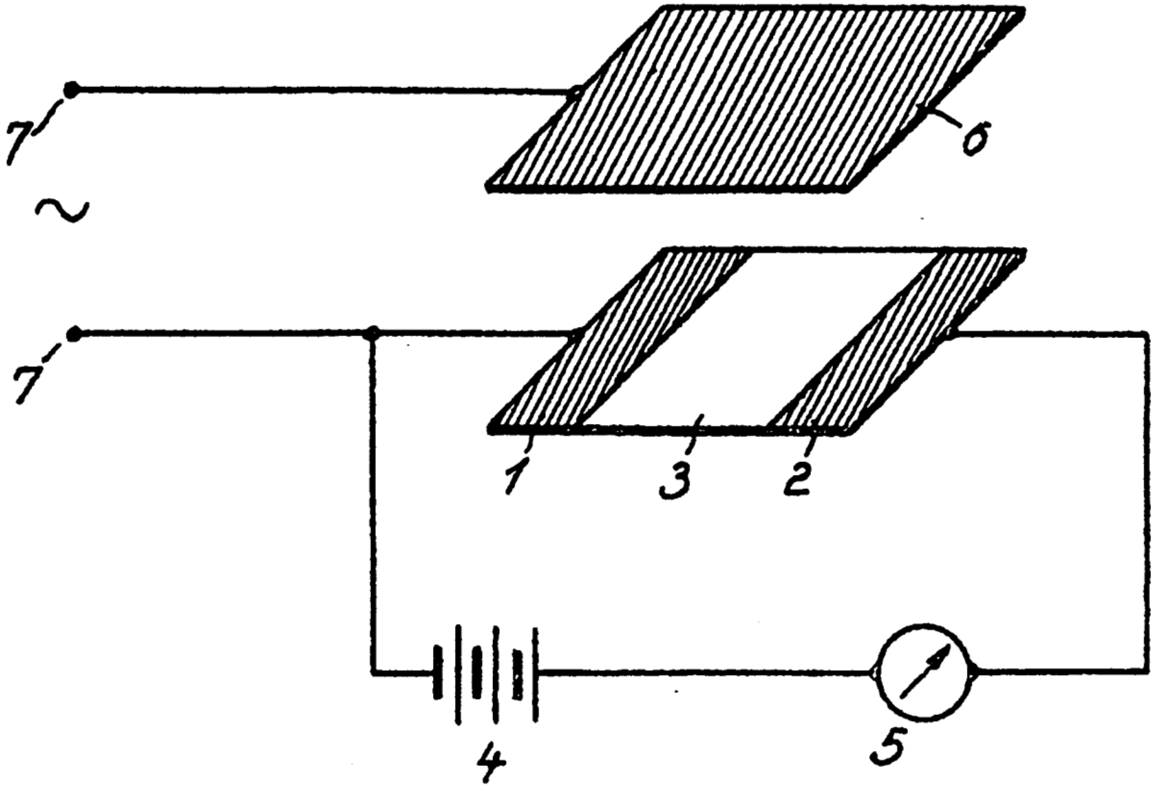


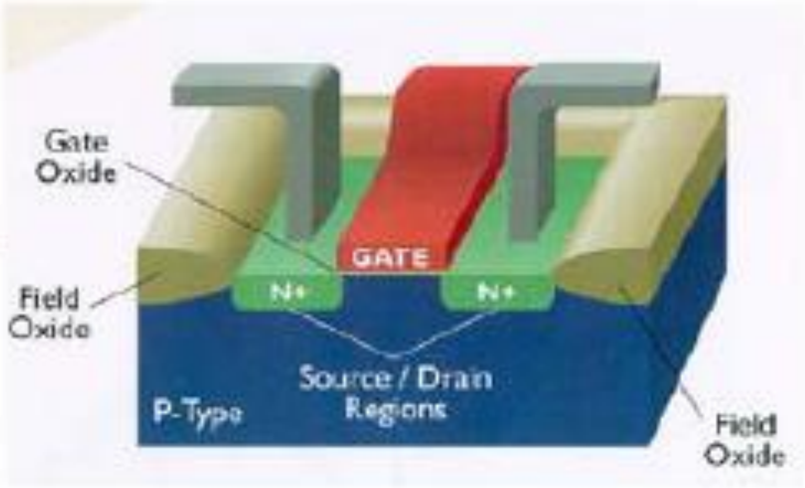
In 1935, a British patent was issued to Oskar Heil.
A working MOSFET was not demonstrated until 1955.

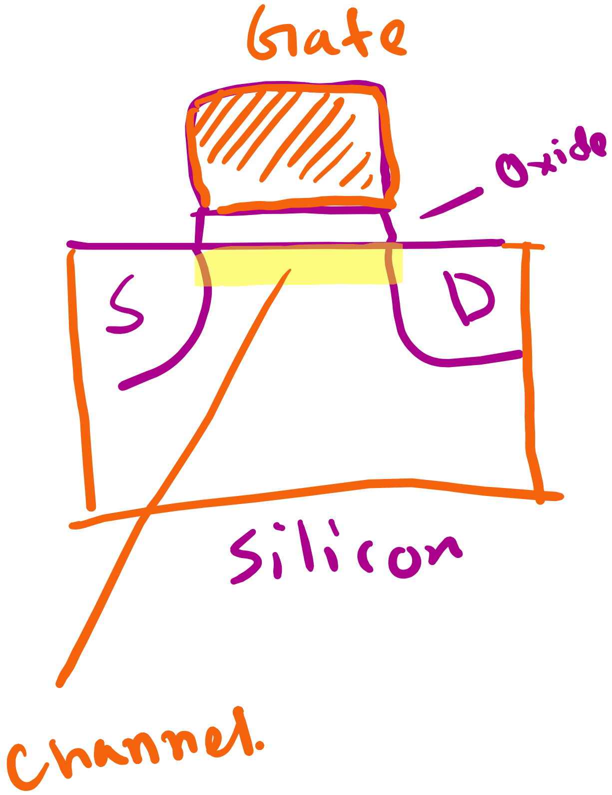
~~Modern~~ Field Effect Transistor (FET)

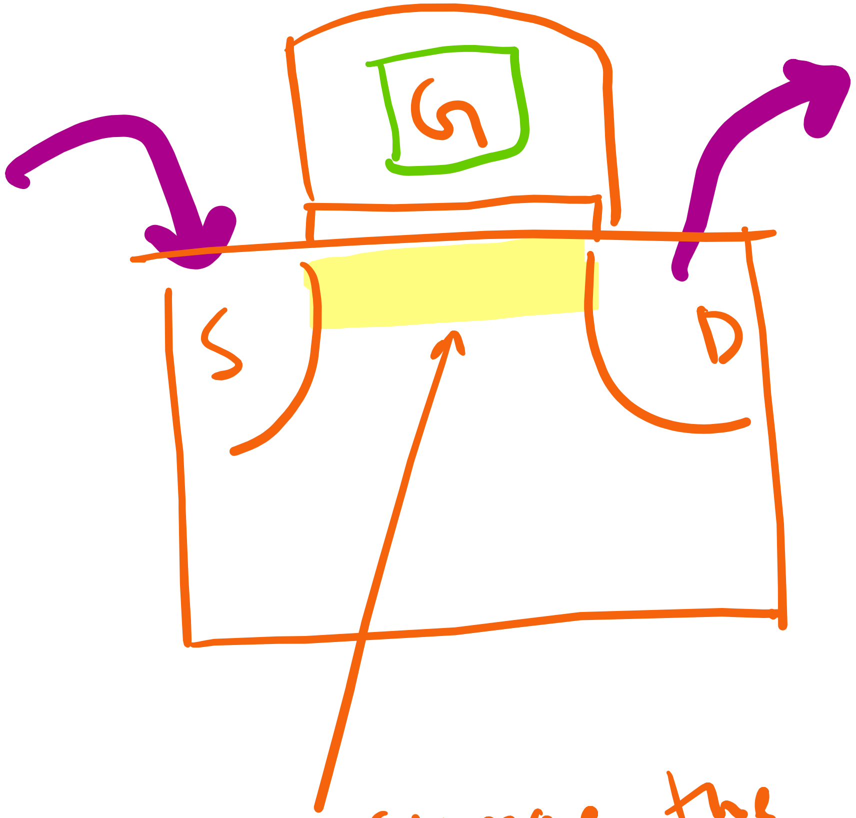
- An electric field is applied normal to the surface of the semiconductor (by applying a voltage to an overlying electrode), to modulate the conductance of the semiconductor

→ Modulate drift current flowing between 2 contacts ("source" and "drain") by varying the voltage on the "gate" electrode

N-channel MOSFET:







change the resistance of the channel.

MOSFET

Metal oxide semiconductor $\boxed{\text{field effect}}$ transistor.

high resistance

↓

no flow

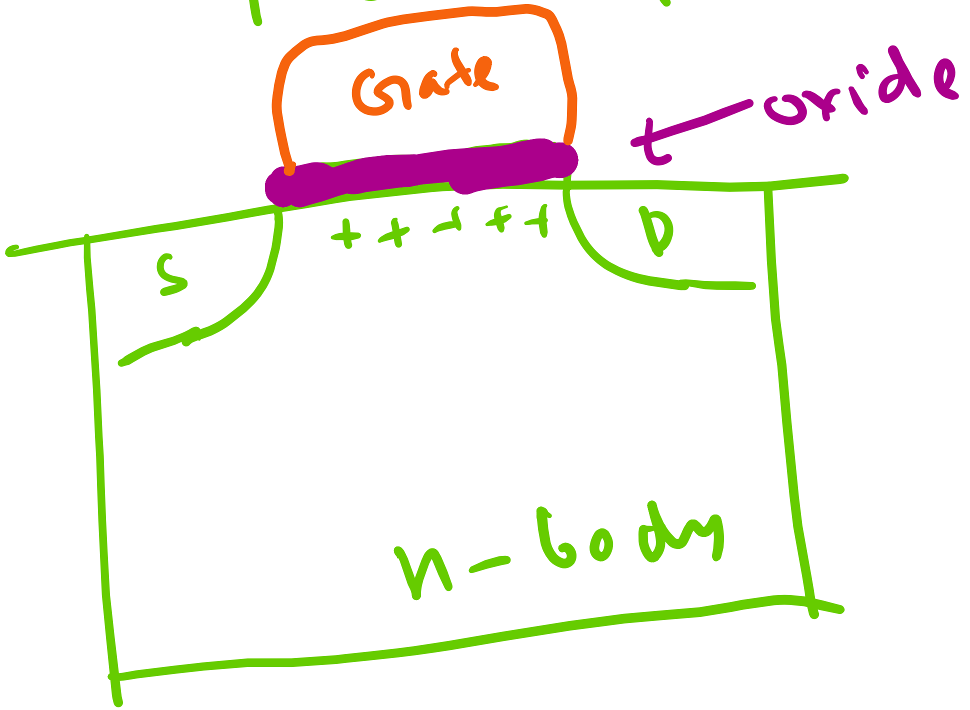
low resistance

↓

high flow

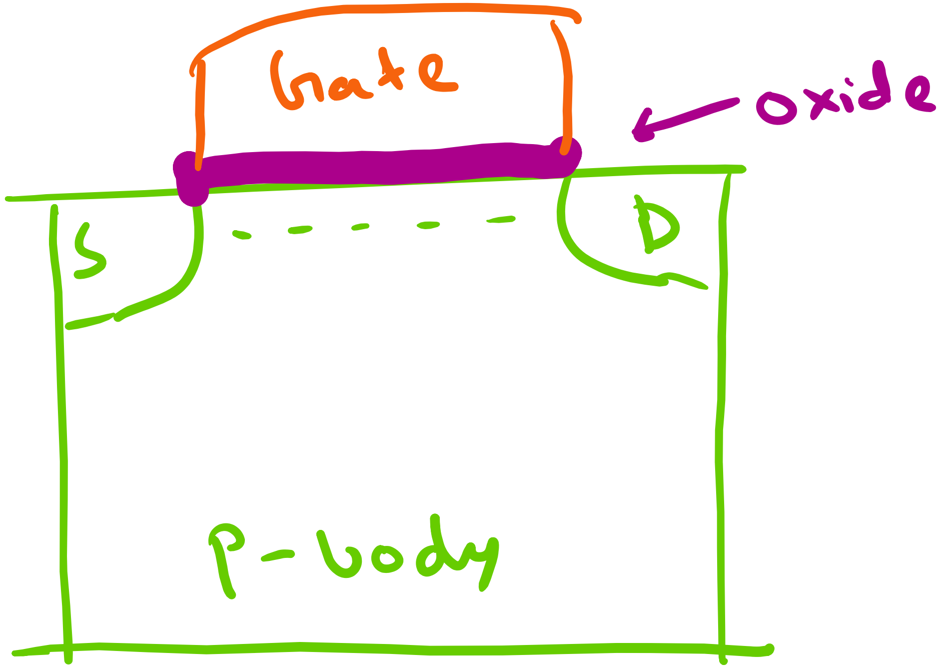
−(Negative) turn ON

P-channel



+(Positive) turn on

N-channel MOSFET.

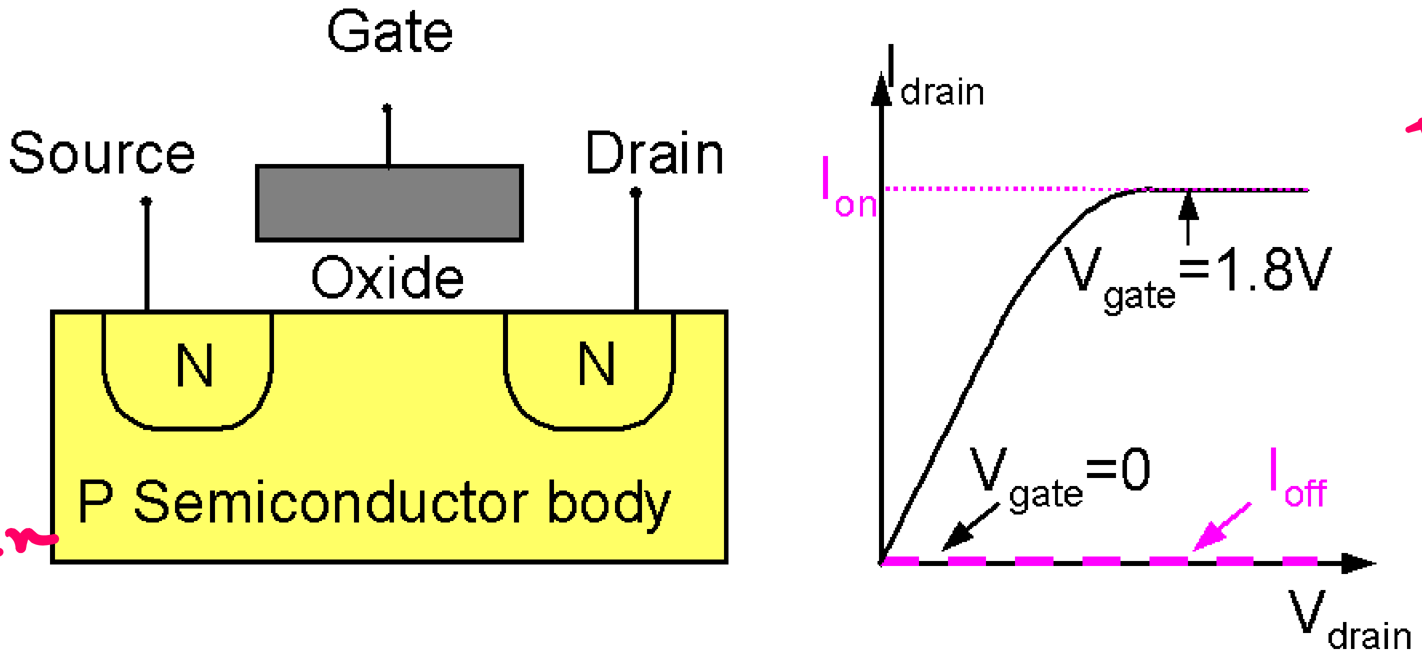


Introduction to the MOSFET

I_{DS}

Current going from drain to source

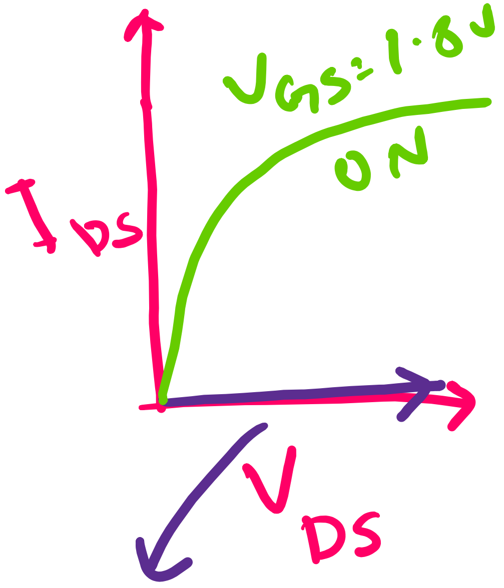
Basic MOSFET structure and IV characteristics



What is desirable: large I_{on}, small I_{off}

V_{DS}

Voltage between drain and source



$V_{GS}=0$

$I_{DS}=0$

OFF state

Introduction to the MOSFET

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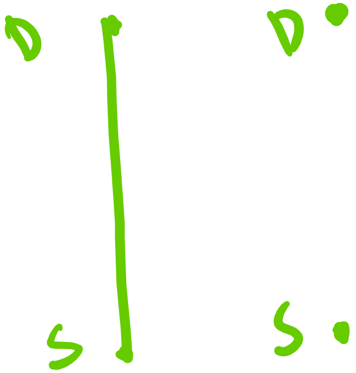
n $\boxed{V_{GS} > V_{TH}}$

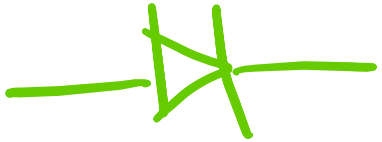
p $\boxed{V_{GS} < V_{TH}}$

ON OFF

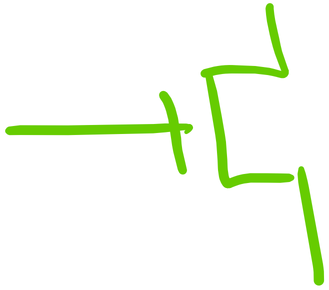


Two ways of representing a MOSFET:

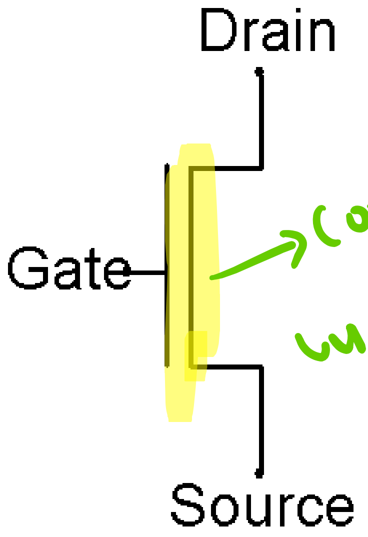
Diode



Transistor.

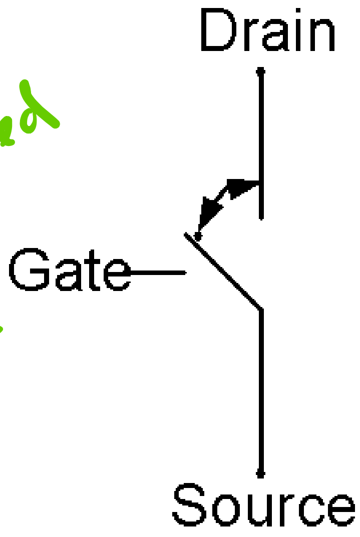


Circuit Symbol



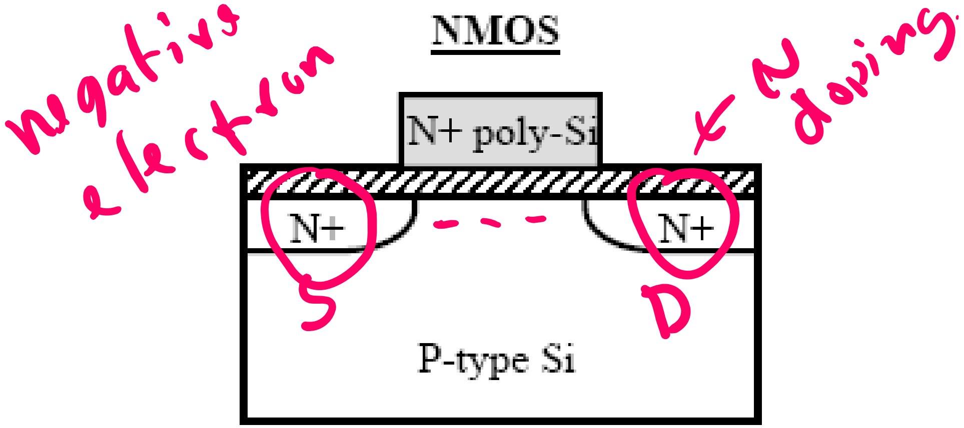
controlled by the gate

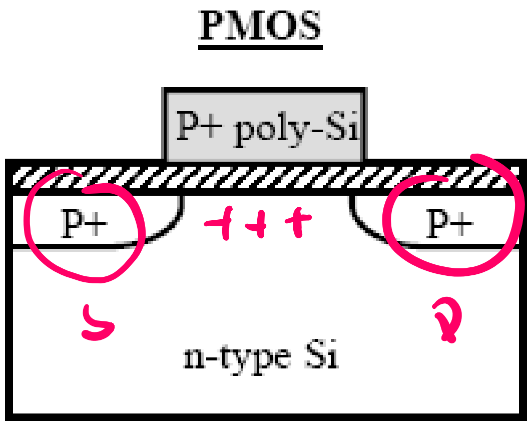
Simple Switch



N-channel = NMOS
P-channel = PMOS

N-channel *vs.* P-channel

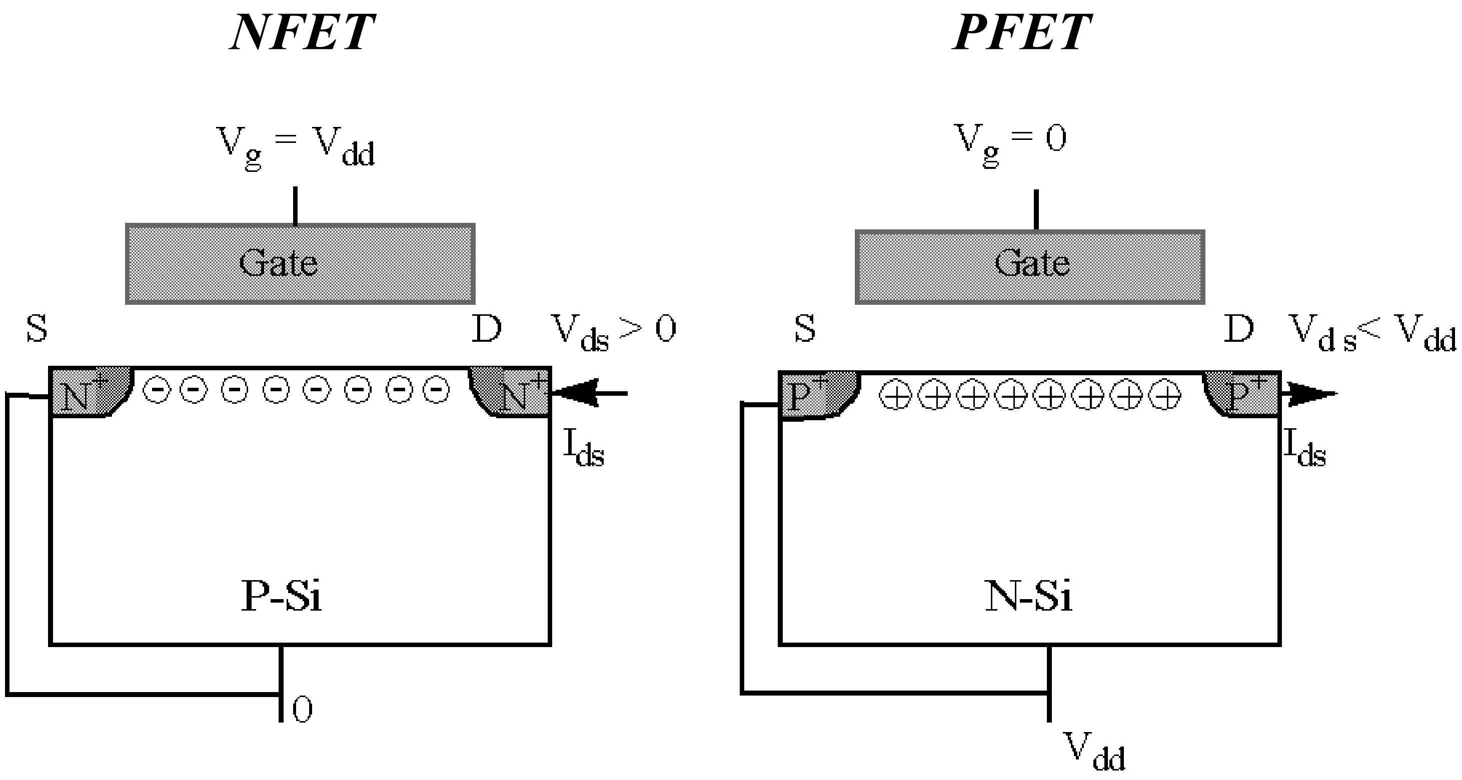




- For current to flow, $V_{GS} > V_T$
- Enhancement mode: $V_T > 0$

- For current to flow, $V_{GS} < V_T$
- Enhancement mode: $V_T < 0$

Complementary MOSFETs (CMOS)



When $V_g = V_{dd}$, the NFET is on and the PFET is off.
When $V_g = 0$, the PFET is on and the NFET is off.

I_D-V_{DS} characteristics expected from a long channel ($\Delta L << L$) MOSFET (n-channel), for various values of V_G

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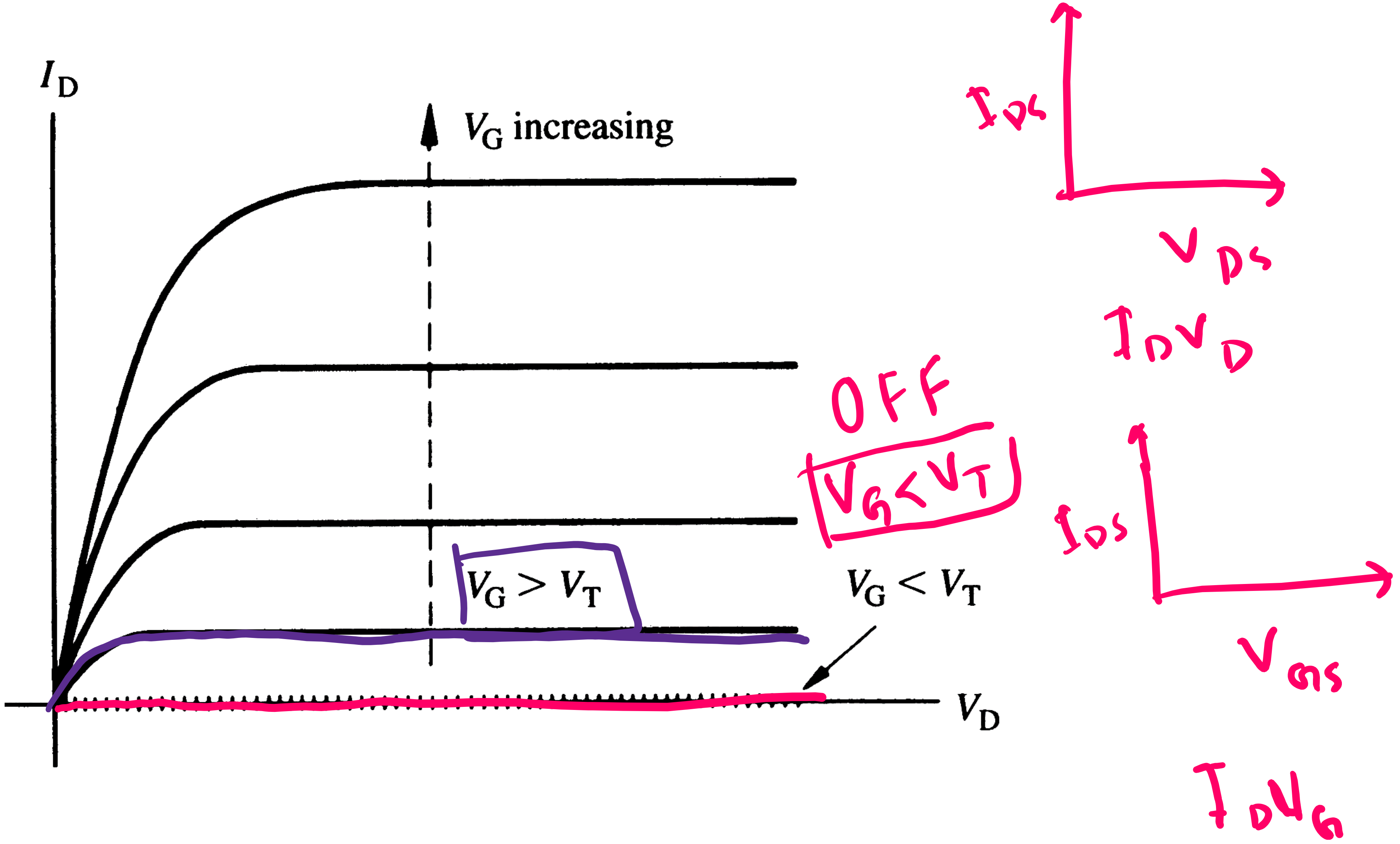


MOSFET current equations

MOSFET Equations

- Linear (ohmic) region:

$$I_D = \boxed{\mu C_{ox}\frac{W}{L}}\left[(V_{GS}-V_{TH})V_{DS}-\frac{1}{2}V_{DS}^2\right]$$

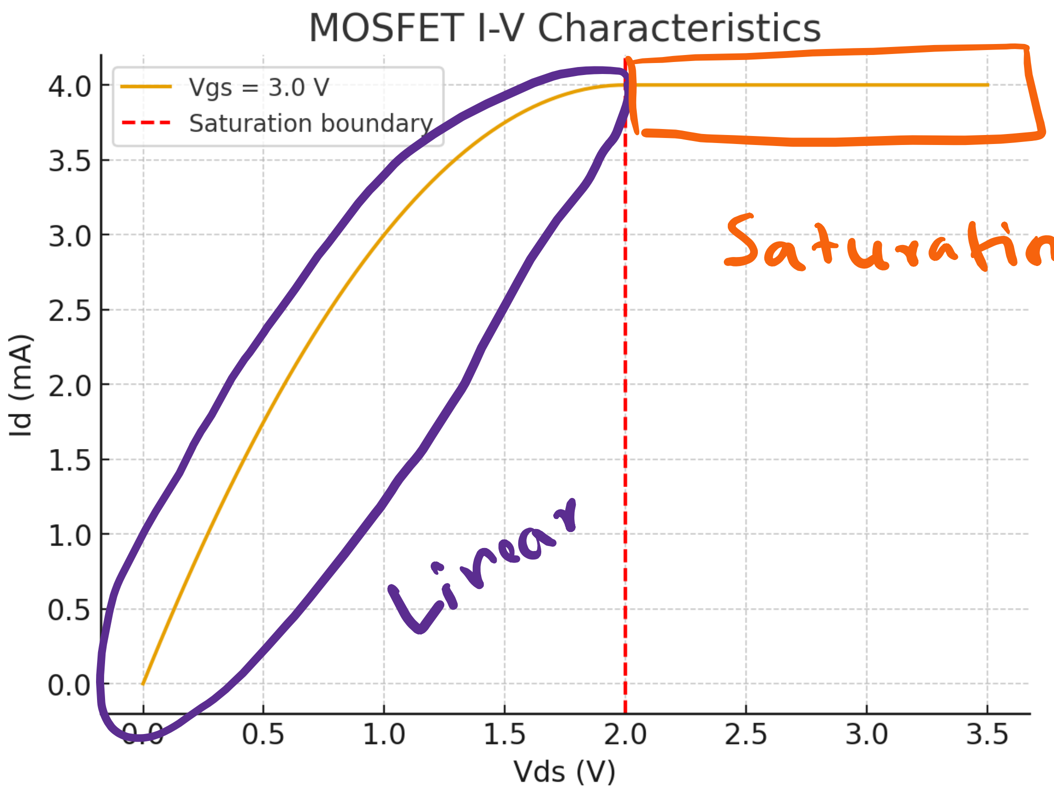
Drain current.

(valid when $V_{DS} < V_{GS} - V_{TH}$)

- Saturation region:

$$I_D = \frac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2$$

(valid when $V_{DS} \geq V_{GS} - V_{TH}$)



Assume Device Parameters

- Mobility–oxide product: $\mu C_{ox} = 200\,\mu A/V^2$
- Width/Length ratio: $W/L = 10$
- Threshold voltage: $V_{TH} = 1.0\,V$

So,

$$k' = \boxed{\mu C_{ox}\frac{W}{L}} = 200 \times 10 = 2000\,\mu A/V^2 = \boxed{2\,mA/V^2}$$

MOSFET Specific.

MOSFET current equations

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MOSFET Equations

- Linear (ohmic) region:

$$I_D = \mu C_{ox} \frac{W}{L} \left[(V_{GS} - V_{TH}) V_{DS} - \tfrac{1}{2} V_{DS}^2 \right]$$

(valid when $V_{DS} < V_{GS} - V_{TH}$)

- Saturation region:

$$I_D = \tfrac{1}{2} \mu C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2$$

(valid when $V_{DS} \geq V_{GS} - V_{TH}$)

Assume Device Parameters

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- Width/Length ratio: $W/L = 10$
- Threshold voltage: $V_{TH} = 1.0\, V$

So,

$$k' = \mu C_{ox} \frac{W}{L} = 200 \times 10 = 2000\, \mu A/V^2 = 2\, mA/V^2$$

Example 1 — Linear region

- $V_{GS} = 3.0\, V$
- $V_{DS} = 0.5\, V$

Check region: $V_{GS} - V_{TH} = 2.0\, V$. Since $V_{DS} = 0.5 < 2.0$, device is in **linear region**.

$$I_D = 2 \left[(2.0)(0.5) - \tfrac{1}{2}(0.5)^2 \right]$$

$$= 2\,[1.0 - 0.125] = 2 \times 0.875 = \boxed{1.75\, mA}$$

$I_D = 1.75\, mA.$

MOSFET current equations

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MOSFET Equations

- Linear (ohmic) region:

$$I_D = \mu C_{ox}\frac{W}{L}\left[(V_{GS}-V_{TH})V_{DS}-\tfrac{1}{2}V_{DS}^2\right]$$

(valid when $V_{DS} < V_{GS} - V_{TH}$)

- Saturation region:

$$\boxed{I_D = \tfrac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2}$$

(valid when $V_{DS} \geq V_{GS} - V_{TH}$)

Assume Device Parameters

- Mobility–oxide product: $\mu C_{ox} = 200\,\mu A/V^2$
- Width/Length ratio: $W/L = 10$
- Threshold voltage: $V_{TH} = 1.0\,V$

So,

$$k' = \mu C_{ox}\frac{W}{L} = 200 \times 10 = 2000\,\mu A/V^2 = 2\,mA/V^2$$

Example 2 — Saturation region

- $V_{GS} = 3.0\,V$
- $V_{DS} = 3.0\,V$

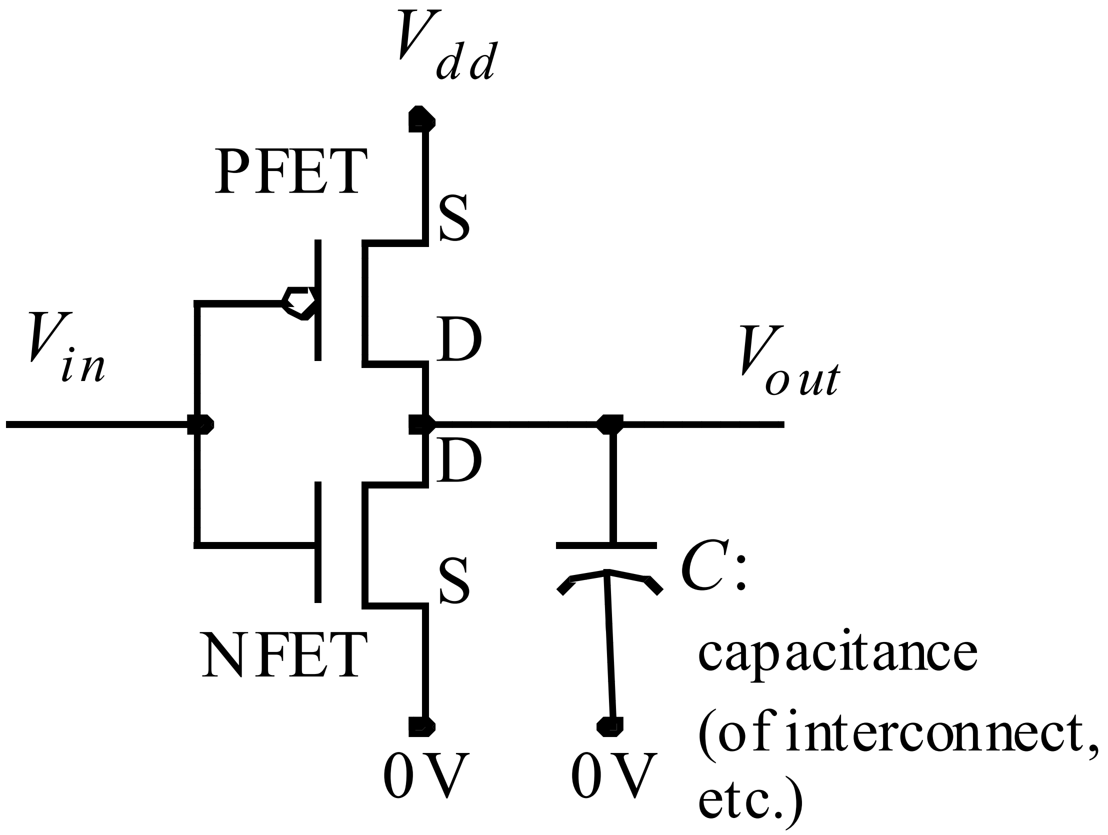
Check region: $V_{GS} - V_{TH} = 2.0\,V$. Since $V_{DS} = 3.0 \geq 2.0$, device is in saturation.

$$I_D = \tfrac{1}{2} \times 2 \times (2.0)^2$$

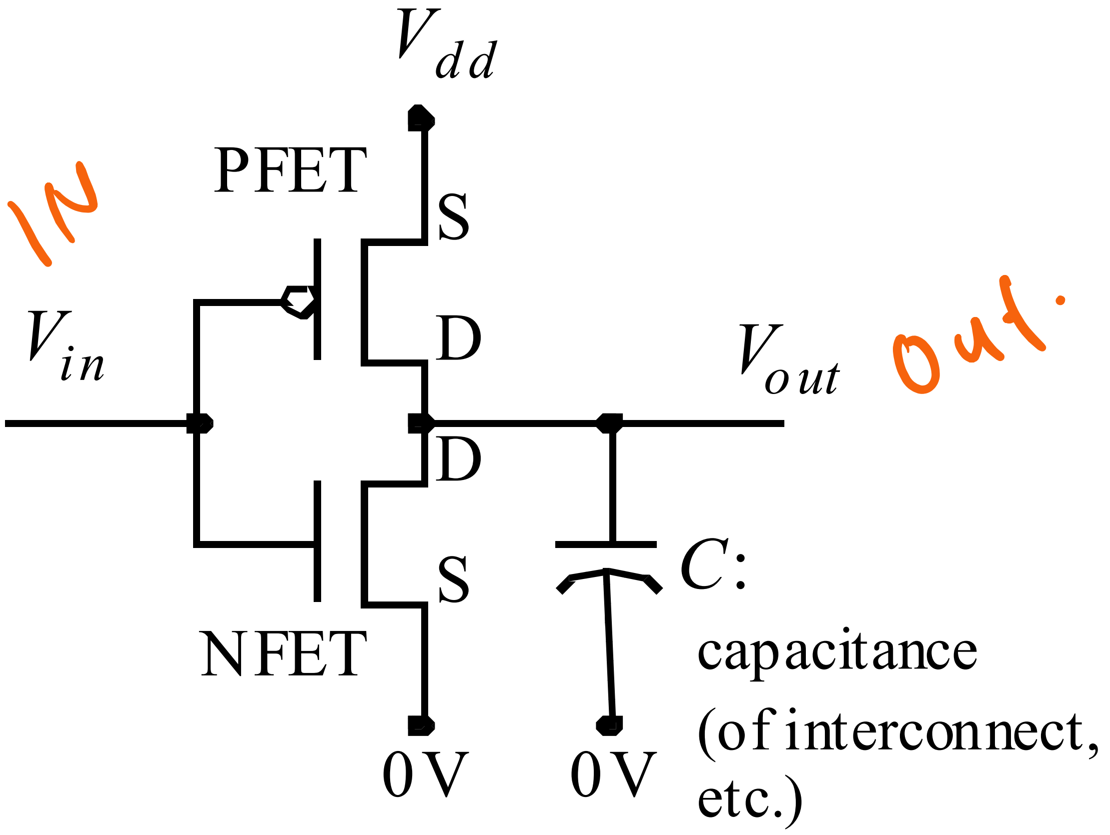
$$\boxed{= 1 \times 4.0 = 4.0\,mA}$$

$I_{Dsat} = 4mA.$

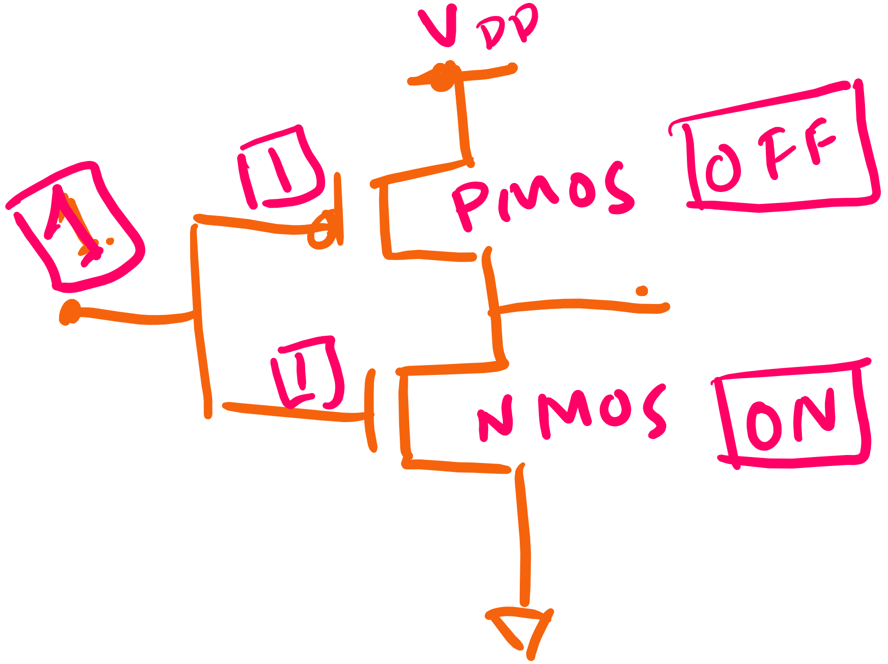
CMOS (Complementary MOS) Inverter

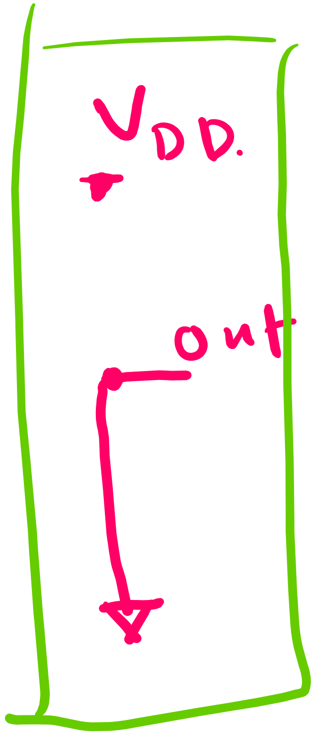


CMOS (Complementary MOS) Inverter

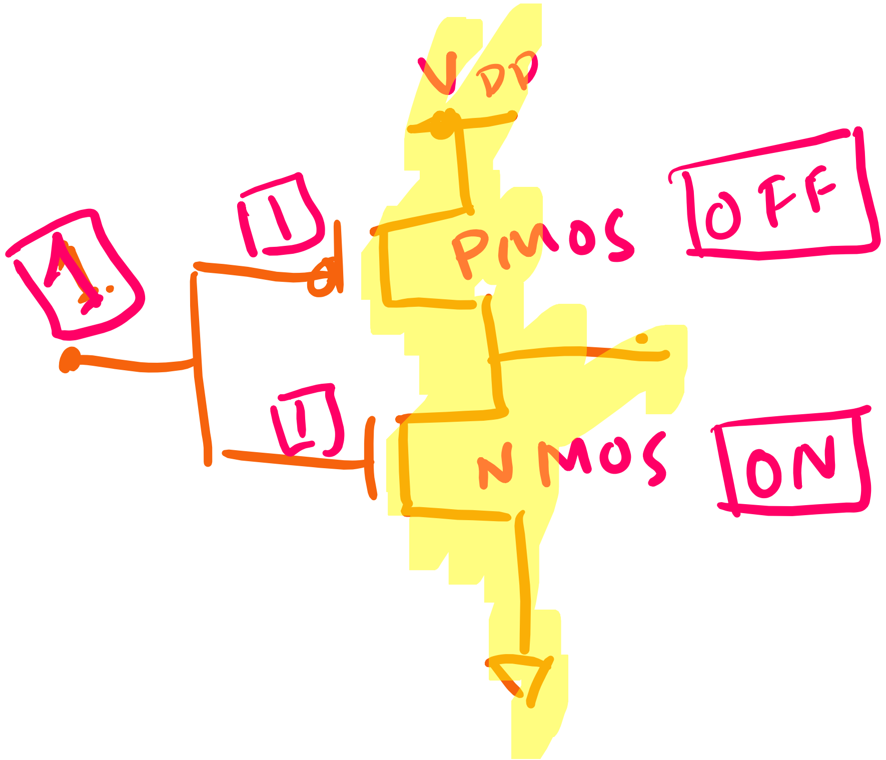


IN	out.
$\boxed{V_{DD}}$	0V
0	



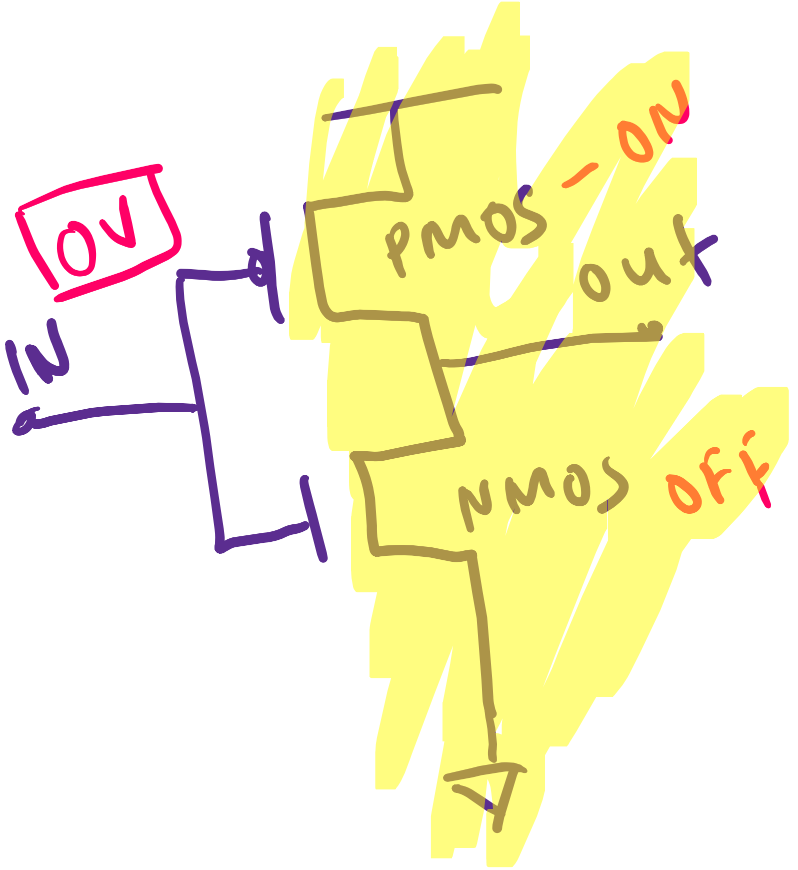


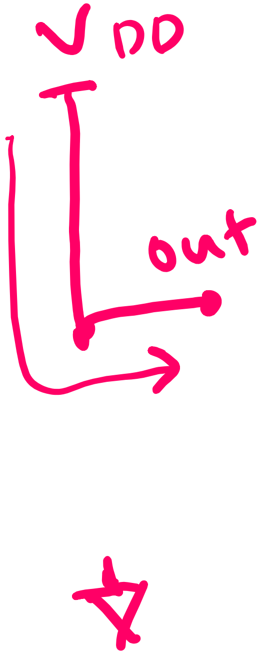
CMOS (Complementary MOS) Inverter



IN	out.
VDD	0V
0	

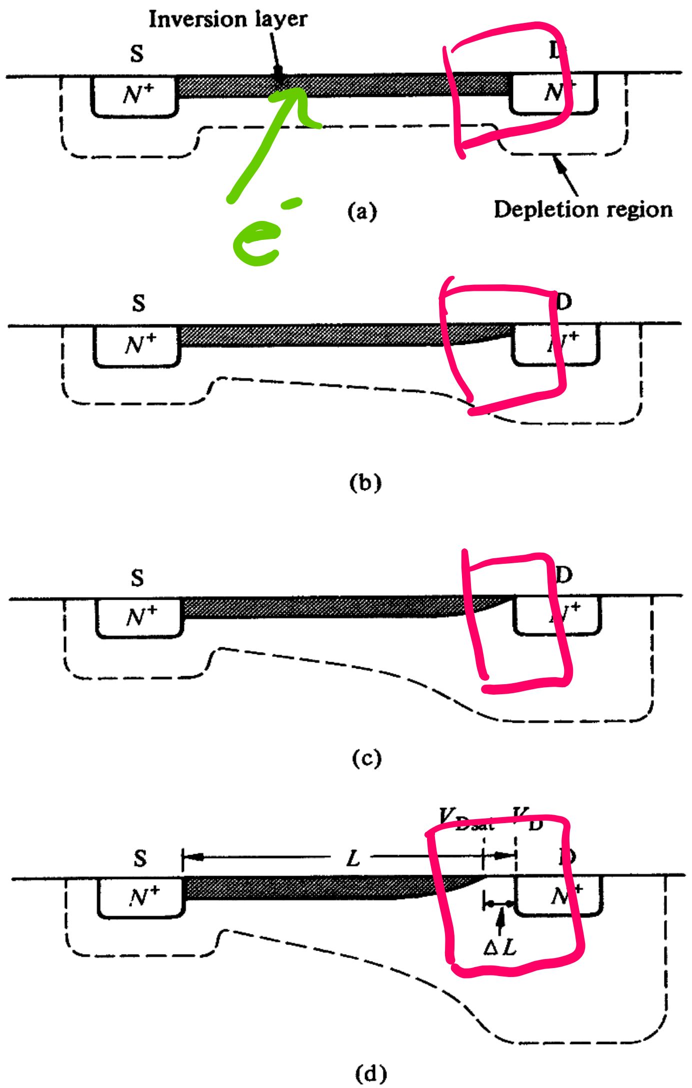
CMOS (Complementary MOS) Inverter

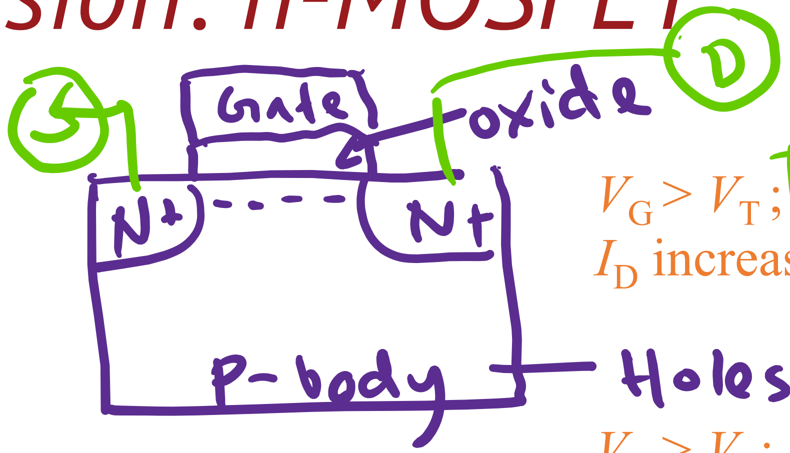




IN	out.
$\boxed{V_{DD}}$	0V
0	V_{DD}

Qualitative discussion: n-MOSFET





V_{DS}

$V_G > V_T$; $V_{DS} \approx 0$
I_D increases with V_{DS}

$V_{DS} = 0$

Holes → majority, electrons - minority

$V_G > V_T$; V_{DS} small, > 0
I_D increases with V_{DS}, but rate of increase decreases.

V_{DS} small

$V_G > V_T$; $V_{DS} \approx$ pinch-off
I_D reaches a saturation value, $I_{D,sat}$
The V_{DS} value is called $V_{DS,sat}$

V_{DS} pinch off

$V_G > V_T$; $V_{DS} > V_{DS,sat}$
I_D does not increase further, saturation region.

$V_{DS} >$ V_{Dsat} pinch off volt

Quantitative I_D-V_{DS} Relationships – 1st attempt "Square Law"

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$V_{GS} - V_{DS} = V_T$

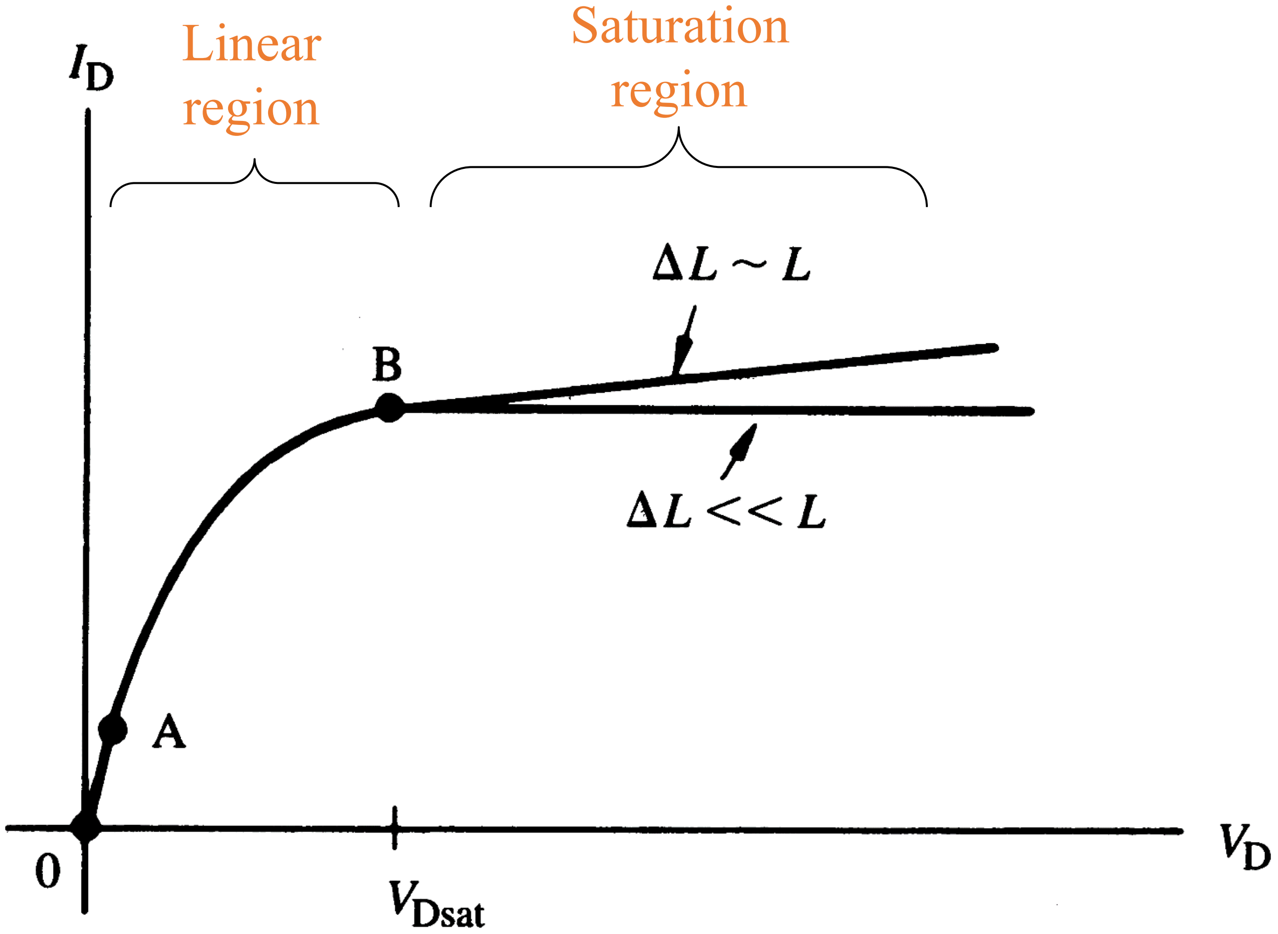
$$I_D = \mu C_{ox}\frac{W}{L}\left[(V_{GS} - V_{TH})V_{DS} - \tfrac{1}{2}V_{DS}^2\right] \qquad 0 < V_{DS} < V_{DS,sat} \;;\quad V_G > V_T$$

I_D will increase as V_{DS} is increased, but when $V_G - V_{DS} = V_T$, pinch-off occurs, and current saturates when V_{DS} is increased further. This value of V_{DS} is called $V_{DS,sat}$. i.e., $V_{DS,sat} = V_G - V_T$ and the current when $V_{DS} = V_{DS,sat}$ is called $I_{DS,sat}$.

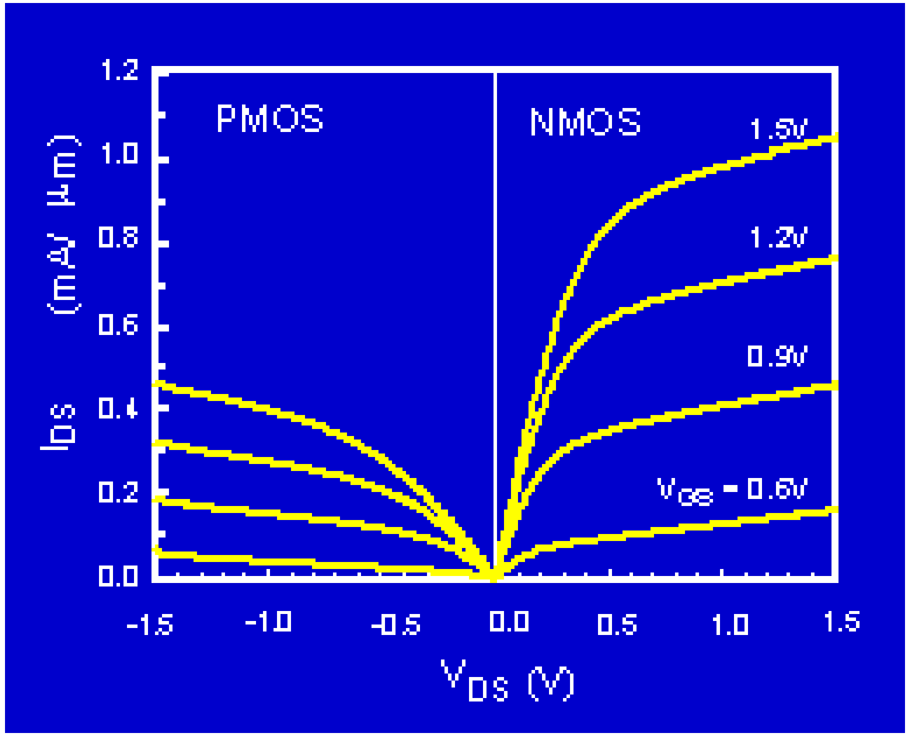
$$I_D = \tfrac{1}{2}\mu C_{ox}\frac{W}{L}(V_{GS} - V_{TH})^2 \qquad V_D > V_{DS,sat} \;;\quad V_G > V_T$$

Here, C_{ox} is the oxide capacitance per unit area, $C_{ox} = \varepsilon_{ox} / x_{ox}$

I_D-V_{DS} characteristics for n-MOSFET

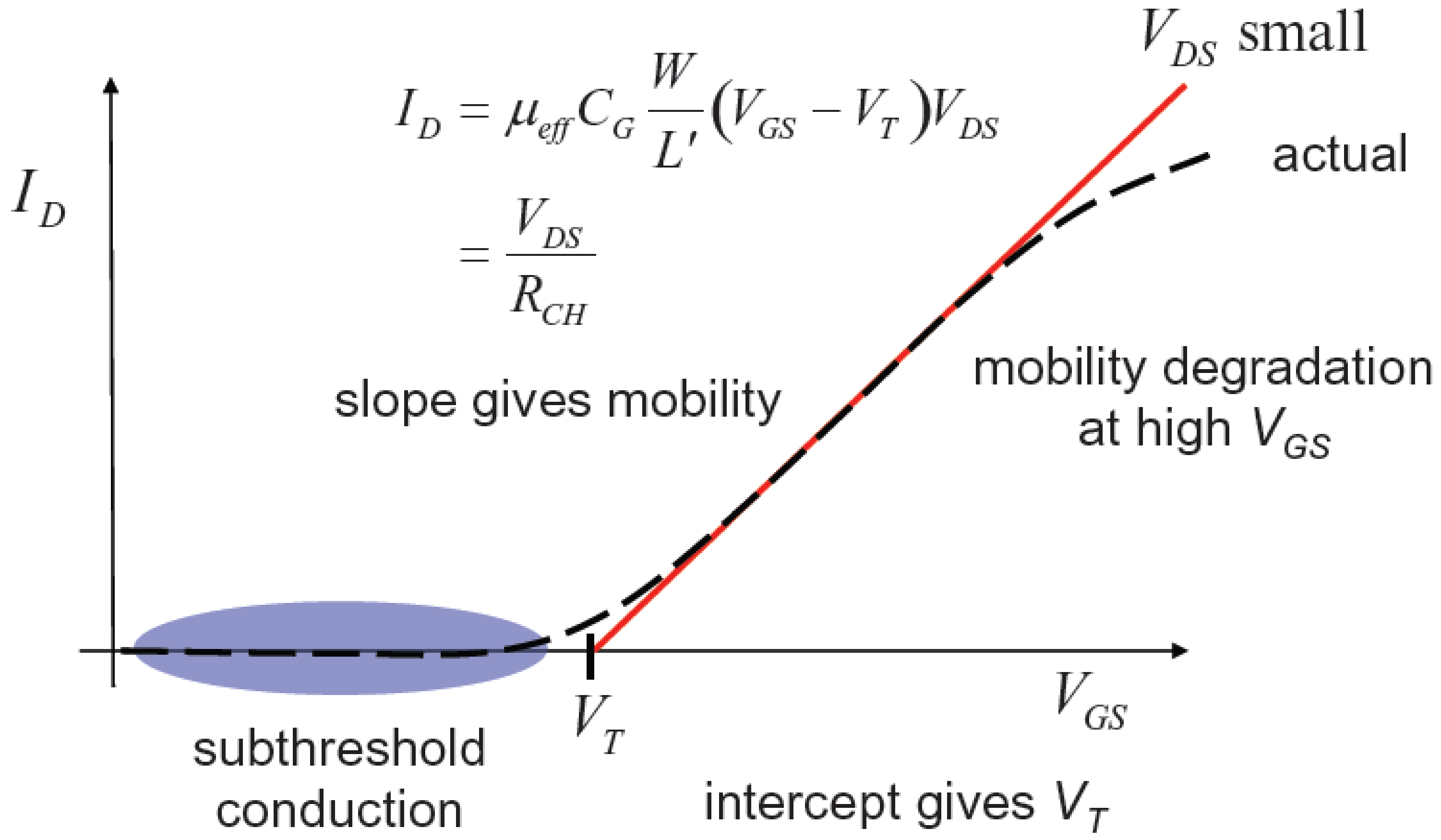


P-MOSFET N-MOSFET IV Characteristics



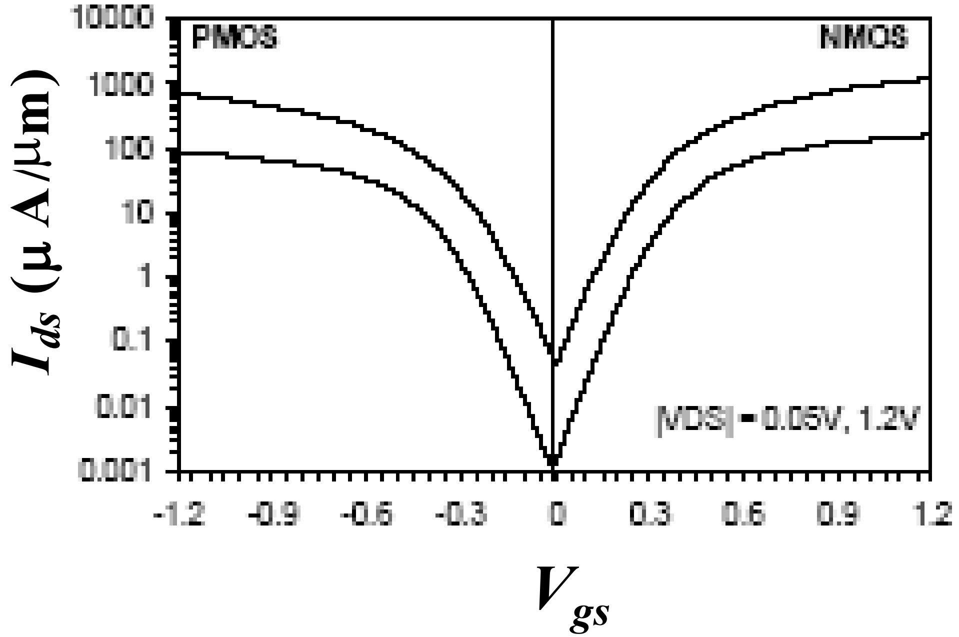
The PMOS IV is qualitatively similar to the NMOS IV, but the current is about half as large. Why?

Threshold and Subthreshold



Subthreshold Current

- The leakage current that flows at $V_g<V_t$ is called the subthreshold current. Previously we had assumed that current is zero, but in reality that's not the case.



Intel, T. Ghani et al., IEDM 2003

90nm technology. Gate length: 45nm for NMOS, 50nm for PMOS

- The current at $V_{gs}=0$ and $V_{ds}=V_{dd}$ is called I_{off}.